cript. NativeScript is released as an Open Source technology, released under Open Source Apache 2 license (https://github.com/NativeScript/NativeScript).

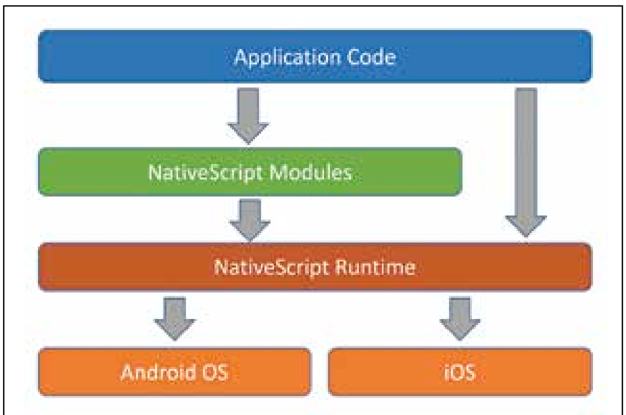
Why NativeScript?

Let's go through some of the salient features of NativeScript:

- **Native UI:** With NativeScript you get Native UI Platform – no Web Views. You define once and NativeScript will adapt and run everywhere.
- **Extensible:** You can easily reuse existing plugins from NPM, Cocoapods (iOS), Gradle (Android) and many NativeScript specific plugins on NPM.
- **Cross-Platform:** With just a single code base you now have the ability to write and deploy native mobile apps to iOS, Android and (soon) Windows.
- **Accelerated Learning Curve:** Since you reuse skills you already know, the learning curve is very minimal.
- **Open Source:** NativeScript is released under Open Source Apache 2 license.
- **Backed by Progress:** NativeScript as a technology is backed by the global leader in application development Progress. The company powers startups and industry titans and has built a community of over 1.9 million developers worldwide.

How NativeScript Works?

NativeScript or {N} as it is popularly addressed, is built from ground up. There are no DOM, no cross compilation and no plugins required. NativeScript is straight up JavaScript and runs as Native App. Under the hood it works over an abstraction. Your JS code is ran against a Virtual Machine packaged during build. NativeScript packages JavaScriptCore VM on iOS and Google V8 VM on Android. It provides 100% access to native APIs. NativeScript utilizes a bridge which is available in the runtime and uses reflection to look up native APIs at execution. During build time, it injects a metadata generated against list of APIs for the platform you build. Here is a block diagram which depicts NativeScript working at a high level:



NativeScript under the hood

Getting Started with NativeScript:

NativeScript provides a Command Line Interface (CLI). With NativeScript CLI you can develop native apps on your development machine for free. You can use text editor of your choice to code. NativeScript as a technology is supported on Windows, Linux & Mac OS. Note that on Windows/Linux you can only create Android packages and not iOS package.

Set up your System:

I will be using a Windows machine as the development platform for the rest of the article. There are certain pre-requisites you will need to install before you begin. They are:

- Node JS
- JDK 8 or later stable official release
- Android SDK

Once the pre-requisites have been installed, open a command prompt and type the following code:

```
npm i -g nativescript
```

The above command installs NativeScript on your machine. You will use NativeScript commands using the shortcut "tns" (which stands for Telerik NativeScript). In order to check if everything has been installed correctly or not, you can execute the following in the command prompt:

```
tns doctor
```

`tns doctor` command checks your pre-requisite and NativeScript installation. If everything is in place, it will state that no issues were detected.

Create a NativeScript Project:

It's time to create a mobile app now. For this exercise I have chosen TypeScript as the language. TypeScript is a Superset of JavaScript which provides certain typed paradigms such as strong typing, Class, Interface etc. Run the following command on a command prompt:

```
tns create MyApp --tsc
```

We create a new app by the name "MyApp" and tell NativeScript to use TypeScript as the language. NativeScript will go ahead and create the project for us.

Add Target Platforms:

Once the project has been created, we need to then add target platforms. Navigate to your project folder in command prompt and execute the following command:

```
cd MyApp
tns platform add android
```

Since I am on a Windows development environment, I can add only Android as a target platform. Whereas if you are on Mac OS, you can add both Android & iOS as target platforms.

Editing NativeScript Projects:

In order to work with NativeScript projects, you can use any editor of your choice. NativeScript is all about XML, JavaScript and CSS files. I will be using one of my favorite open source editors from Microsoft called "Visual Studio Code". It is a fantastic editor if you are working heavily on JavaScript/TypeScript based projects. Next, let's take a look at the project structure.

NativeScript Project Structure:

In Visual Studio Code, I have opened our project folder. As